INDIAN INSTITUTE OF TECHNOLOGY PATNA

EC3202: DIGITAL SIGNAL PROCESSING LAB

Experiment No: 06

Topic: Filter Design using Windowing Techniques

Submitted By:

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Aim: Design a Low pass filter using Hamming and Blackman window.

$$z = \frac{\sin(2\pi f c p)}{\pi p}, \text{ where } p = n - \frac{(m-1)}{2}$$

- Consider a truncated sinc function z with angle p as given.
- Plot its truncated unit sample response.
- Plot Frequency response of the truncated sinc function
- Plot the Hamming window function.
- Plot the multiplication of Hamming window and sinc function.
- Frequency response of the windowed sinc function.
- Frequency response of the windowed sinc function in dB.

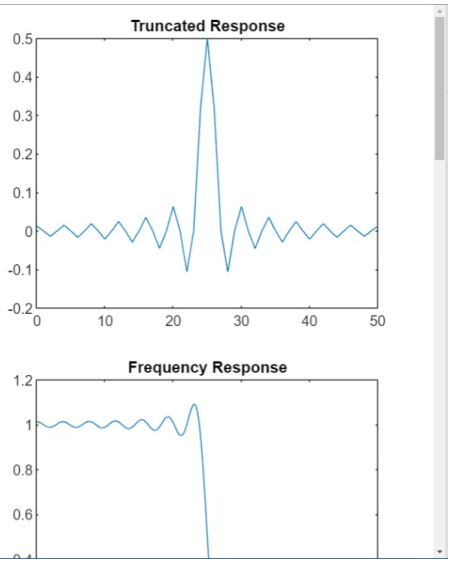
MATLAB CODE AND RESULT :

```
M =51;
fc =0.25;
n = 0:M-1;
p = n-(M-1)/2;
z = sin(2*pi*fc*p)./(pi*p);
z((M+1)/2) = 2*fc;
figure;
plot(n,z);
title("Truncated Response")
```

```
[Z,w] = freqz(z,1,1024);
figure;
plot(w/pi,abs(Z));
title("Frequency Response")
```

```
w = hamming(M);
figure;
plot(w);
title("Hamming of M");
```

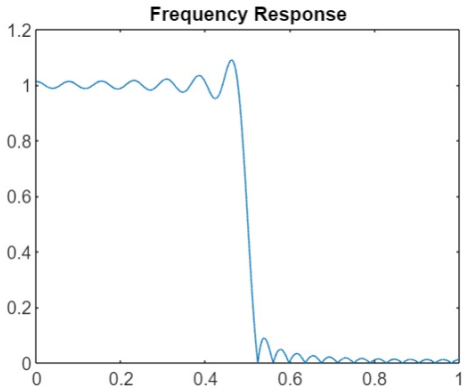
```
x = z.*w';
figure;
plot(x);
```

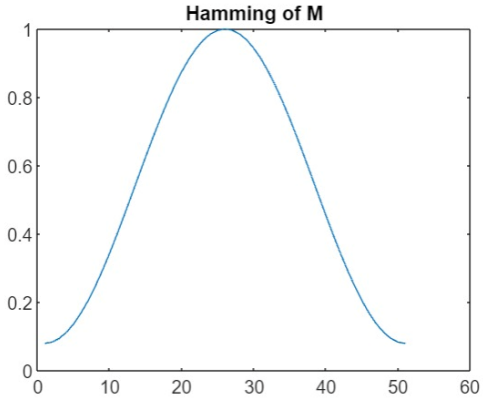


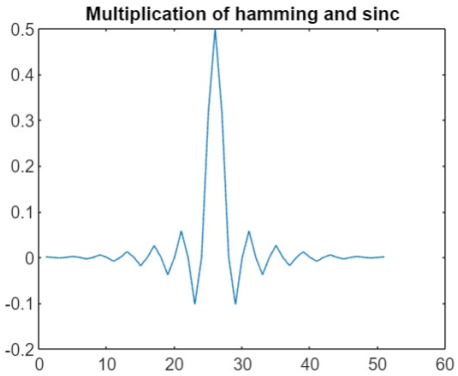

```
x = z.*w';
figure;
plot(x);
title("Multiplication of hamming and sinc");
```

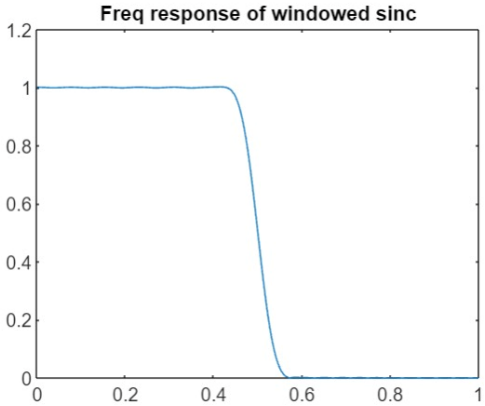
```
[H,h] = freqz(x,1,1024);
figure;
plot(h/pi,abs(H));
title("Freq response of windowed sinc");
```

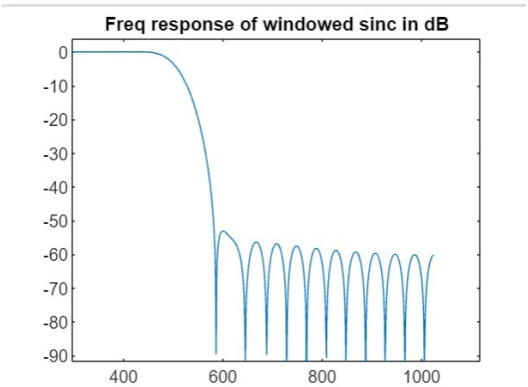
```
plot(20*log10(abs(H)));
title("Freq response of windowed sinc in dB");
```

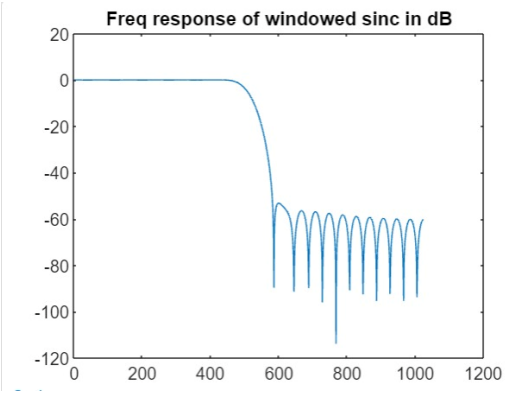

Frequency Response
1.2
1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1

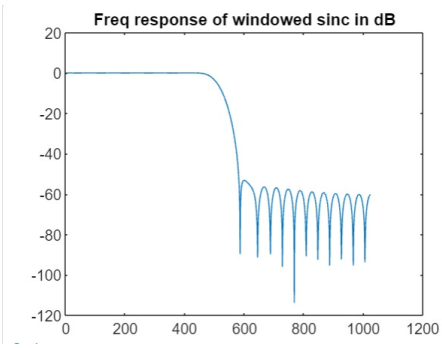

Hamming of M
1
0.8
0.6
0.4
0.2
0
0
10
20
30
40
50
60


Multiplication of hamming and sinc
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
0
10
20
30
40
50
60


Freq response of windowed sinc
1.2
1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1


Freq response of windowed sinc in dB
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
400
600
800
1000


Freq response of windowed sinc in dB
20
0
-20
-40
-60
-80
-100
-120
0
200
400
600
800
1000
1200


Freq response of windowed sinc in dB
20
0
-20
-40
-60
-80
-100
-120
0
200
400
600
800
1000
1200

Conclusion : In this experiment, a low-pass FIR filter was designed using the windowing method with Hamming and Blackman windows**.** The ideal sinc function was truncated and multiplied with the window functions to obtain the filter response. From the frequency response plots, it was observed that the Blackman window provides better stopband attenuation, while the Hamming window has a slightly narrower transition band. Thus, windowing helps control the filter characteristics effectively.